Lab 7b Report Microprocessor

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1 Tone Generation and Keypad Interfacing

1.1 Objective

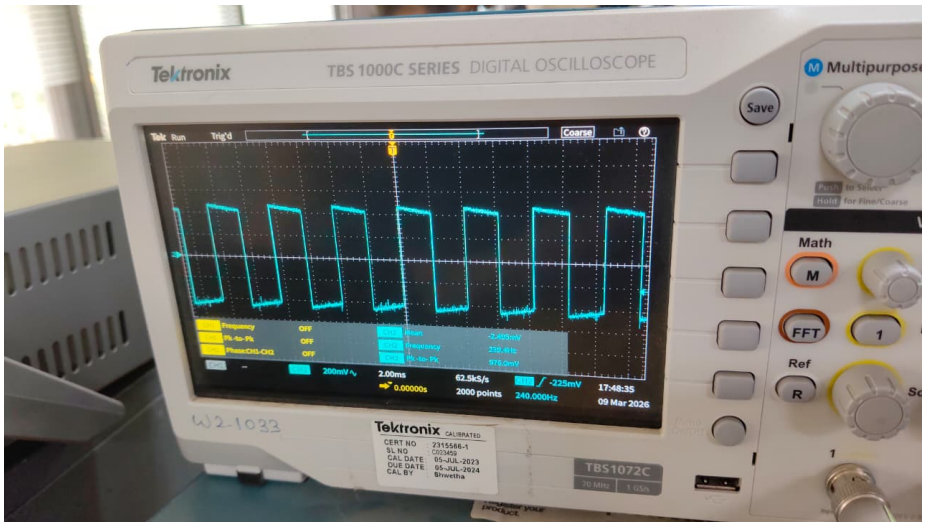
To design and implement a system using the Pt-51 microcontroller board that interfaces with a keypad and an LCD to:

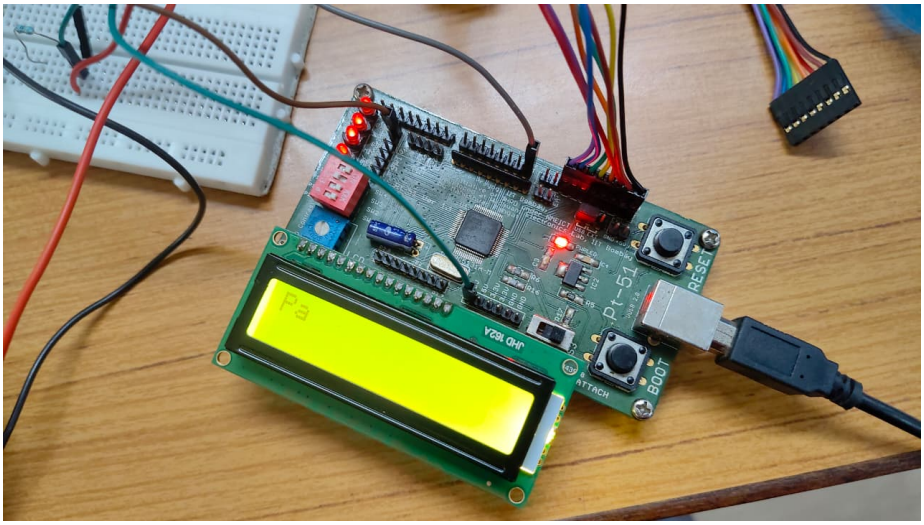
- Generate specific musical notes (square waveforms) through a speaker based on keypad input
- Display the corresponding name of the musical note (e.g., Sa, Re, Ga) on the LCD
- Maintain the tone and display until a new valid key is pressed

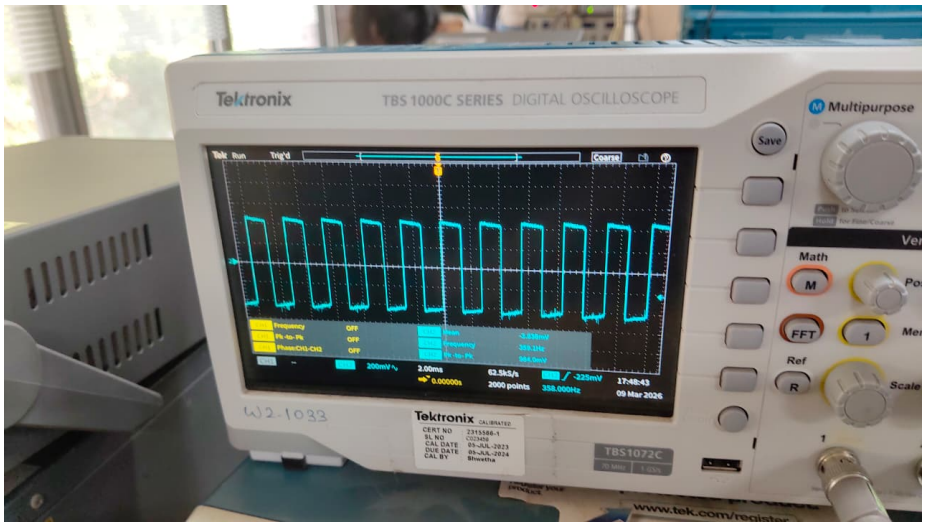
1.2 Observation

During the experiment, the following mappings were verified. A "Success" in the output columns indicates that the frequency was audible and the LCD string matched the table

Key	Note	Frequency (Hz)
1	Sa (low)	240
2	Re	270
3	Ga	300
4	Ma	320
5	Pa	360
6	Dha	400
7	Ni	450
8	Sa (high)	480
9	Silence	0


Tektronix
TBS 1000C SERIES DIGITAL OSCILLOSCOPE
Save
Multipurpose
Math
M
FFT
1
Ref
R
Coarse
200mV
2.00ms
62.5kS/s
2000 points
-225mV
17:48:35
09 Mar 2026
240.000Hz
Tektronix
TBS1072C


JHD 162A
ATTACH
BOOT
RESET


Tektronix
TBS 1000C SERIES DIGITAL OSCILLOSCOPE
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Tektronix
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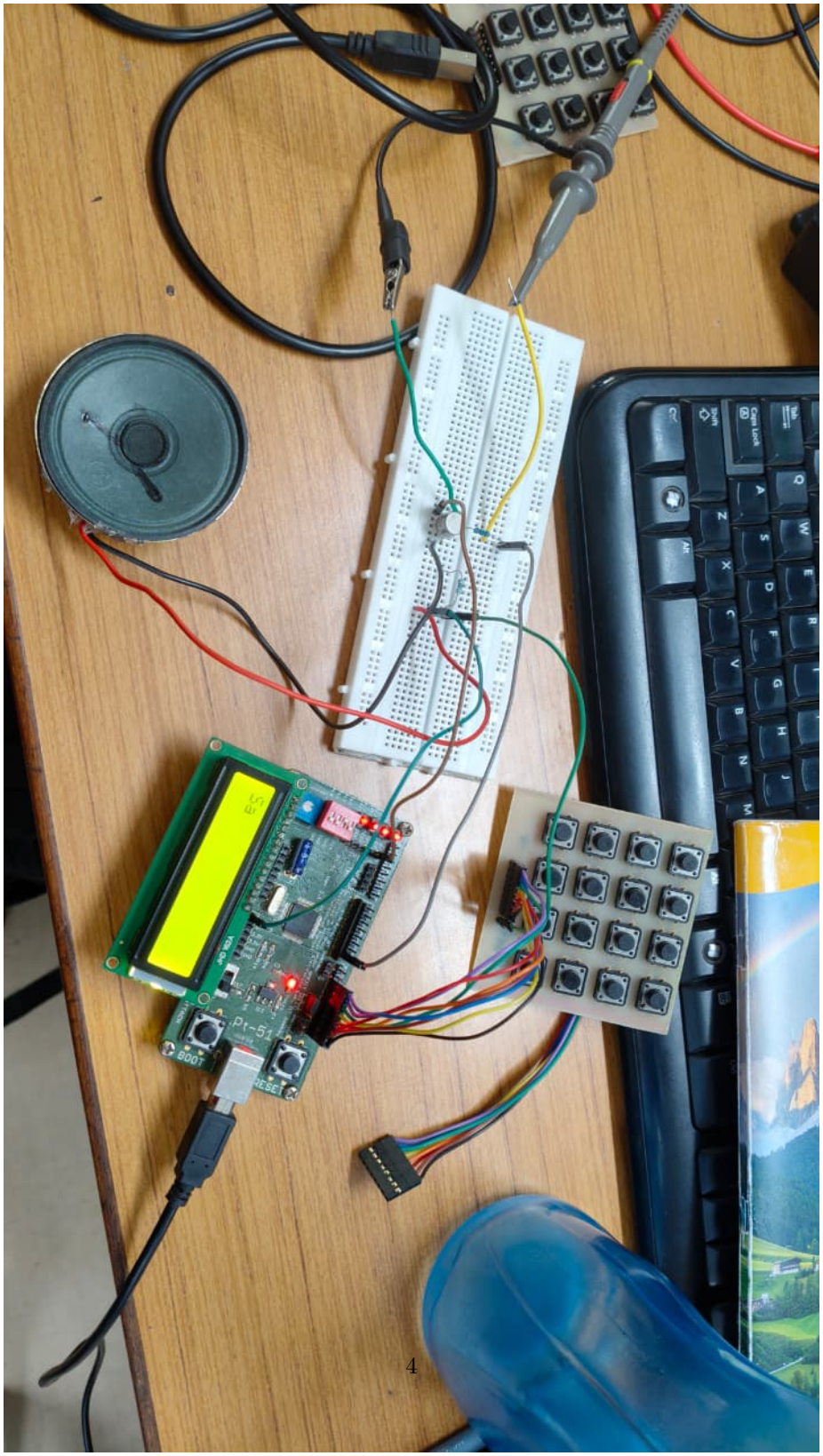
Pt-51
BOOT
RESET

1.3 Conclusion

The experiment successfully demonstrated the integration of input peripherals with output peripherals:

- By calculating the appropriate time delays for the Pt-51 timers, we were able to generate square waves corresponding to the frequencies of the musical scale
- The implementation correctly held the state of the note and the LCD message until a hardware interrupt event detected a new key press
- The logic was verified against the provided Keypad.hex file, ensuring that the hardware mapping and software frequency calculations were accurate